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Climate risks in motorsport: setting boundary conditions in Formula 1

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ABSTRACT

Research question: The global nature of Formula 1 racing exposes the sport to diverse climate risks. We describe the process for setting evidence-based boundary conditions for climate-related hazards in motorsport, and then apply them to Formula 1.

Research methods: We assess risks for 25 Formula 1 racing circuits, including 24 venues in the 2024 racing schedule and one potential future site, by analysing historical weather data, air quality records, and flood risk projections to identify climate-related hazards that could impact race events.

Results and findings: The boundaries proposed herein are new and require further refinement; however, they allow us to draw some initial conclusions about the vulnerabilities of the F1 series in its current format. Our findings show that extreme heat is the most pervasive threat, affecting 19 of the 25 circuits. Flooding poses risks to 18 locations, poor air quality could compromise 10 sites, and heavy rainfall threatens seven venues. Singapore and Qatar emerged as the most vulnerable locations, while Austria shows lower climate risk profiles.

Implications: These results underscore the need for officials to consider adaptations to Formula 1's race calendar and operations to address evolving environmental challenges. This study provides a foundation for developing climate adaptation plans in motorsport, potentially influencing practices in the wider automotive industry. As climate change intensifies, Formula 1's response to these challenges may set precedents for the global sport and tourism industries facing similar environmental pressures.

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Introduction

In 2023, the Formula 1 (F1) grid experienced unprecedented climate challenges. In May, the Emilia Romagna Grand Prix race was cancelled due to flooding in the region resulting from an unseasonable torrential downpour (Lukiv, 2023); more than 200 ml of rain fell in under 48 h (Valente et al., 2025). Months later, the high temperatures at the Qatar Grand Prix evoked strong criticism from drivers, with Esteban Ocon vomiting in his helmet in the 15th and 16th laps as ambient temperatures on the track exceeded 36 degrees Celsius, with high humidity, leading to cockpit temperatures approaching 50 degrees Celsius (Joseph, 2023). George Russell, who competes for Mercedes and serves as the director of the Grand Prix Drivers' Association, told the media in a post-race interview that the race conditions were 'absolutely brutal' and 'beyond the limit of what was acceptable for driving' (Suttill et al., 2023, paras. 1, 4). While neither of these events has been specifically attributed to climate change, the severity of the impacts on the F1 race organisers and drivers underscores the need for more research into the potential climate impacts on motorsport in the future so that risks can be estimated and mitigated.

Climate hazards, such as heat waves, wildfires, storms, floods, and droughts, are associated with dramatic consequences for human health and physical activities (Bernard et al., 2021). For sport events, which rely on stable environmental conditions for safe conditions of play (Orr & Inoue, 2019), the worsening of climate change could result in hazards which bring frequent and severe disruptions (Ross & Orr, 2022). By 2080, only eight of 21 cities that previously hosted the Olympic Winter Games could be suitable to host due to hotter winters and decreased snow (Scott et al., 2015). On the summer side, a study of heat risks in future Olympic Summer Games showed that by 2085, only 33 of 643 cities with sufficient infrastructure to host the Games would have low-risk climate conditions (Smith et al., 2016). Sport-specific risks caused by climate change have direct consequences on athletes, spectators, and officials (Schneider & Mücke, 2021) and affect the provision of sport facilities (e.g. race-tracks). Taken together, these issues suggest a need to evaluate future climate risks.

The consequences of anthropogenic climate change are becoming a barrier to sport. To understand the potential impacts of climate change in a sport, it is first necessary to set boundary conditions for what might be considered safe competitive conditions (Ross & Orr, 2022). In 2023, The Sport Ecology Group was approached by McLaren Racing requesting research into the climate risks affecting the sites they race. This project aimed to generate an initial set of boundary conditions for motorsport, taking into consideration the available data and extant evidence on various climate hazards' impacts to human health, sport performance, facilities, and sport business (e.g. ticket sales, media rights, insurance costs) – four areas of concern identified together with McLaren Racing. We apply these boundary conditions to identify potential impacts on 25 current or proposed F1 circuits and evaluate their susceptibility to climate-related hazards.

Climate vulnerability in sport

Sport ecology scholars have investigated the bi-directional relationship between sport and the environment, exploring the effects of sport on the environment and vice versa

(McCullough et al., 2020). Although the literature on the intersection of sport and the environment is steadily growing, little is known about the effects of climate change on sport events and elite-level performance (Orr et al., 2022). Determining the current impacts of climate change is challenging due to the lack of attribution studies, and the unpredictable nature of environmental hazards, thus making it also difficult to predict the future. Sport managers and personnel involved in staging sport events could benefit from understanding the range of risks that may impact events, such as extreme heat, storms, and UV radiation (Ross & Orr, 2022). Such knowledge is critical not only from a strategic planning standpoint but also for the safety of all involved.

In climate risk assessment, it is critical to determine potential negative impacts. Estimates of negative impacts are based on the scope and magnitude of the specific hazard condition (Hinkel, 2011). The Intergovernmental Panel on Climate Change proposes two related concepts for measuring potential climate impacts: exposure and sensitivity. First, *exposure* focuses on the specific hazard variables that may negatively impact the observational unit (in this case, a F1 race) and addresses the likelihood the hazard will occur. Second, *sensitivity* centres on the resilience of the observational unit if hazardous conditions are present. In other words, how severe would the consequences be if the hazard were to occur?

When considering the complex operational requirements associated with staging major events, sport organisations may benefit from more narrow, specialized approaches to assess climate threats. For instance, Orr and Inoue's (2019) Climate Vulnerability of Sport Organizations (CVSO) framework can be applied to better understand how climate change impacts sport organisations and their capacity to respond to these challenges. Climate impact on organisations (CIO) focuses on the potential adverse effects of climate change on sport organisations, while organisational climate capacity (OCC) examines their ability to adapt to and manage these impacts. This dual focus is particularly relevant for F1, as the sport faces both immediate operational challenges from extreme weather events and longer-term strategic decisions about race calendars and venue selection. While the focus of these concepts is a singular organization, there are several organizations involved in the F1 series (e.g. the promoter, the teams, F1, the FIA, the media), and therefore, each organization involved might concern themselves with the potential impacts of climate hazards on their involvement in the event, and how they can individually and collectively respond. Indeed, previous research has applied the concepts to groups of organizations: for example, Ross and Orr (2022) applied it to mega-events, which involve several organizations including host committees, the International Olympic Committee, participating National Olympic Committees, international sport federations, media, and more.

The temporal dimension of climate risk manifests uniquely in F1, as the sport's year-round calendar requires simultaneously adapting to both seasonal weather patterns and extreme weather events across multiple climate zones. These unique features create unprecedented challenges in risk assessment and mitigation strategies compared to sports with more geographically or seasonally constrained schedules. Additionally, F1's spatial dimension has become increasingly complex as it expands into new markets, introducing new environmental challenges that must be considered alongside sporting and business factors.

The concept of adaptive capacity, drawn from broader climate change literature, offers a framework for understanding F1's potential responses to climate change. Recent studies

have highlighted the importance of considering multiple stakeholder perspectives when assessing climate vulnerability in sport, as each group may experience different impacts from climate hazards and may have different capacities to adapt (Schneider & Mücke, 2021). In F1, this multi-stakeholder lens is particularly crucial given the complex ecosystem of drivers, race organisers, broadcasters, and spectators, each with distinct vulnerabilities. This multidimensional understanding of climate vulnerability provides a robust theoretical foundation for examining F1's unique position at the intersection of global sport and climate change.

Context

F1 racing represents the pinnacle of global motorsport, and the series has experienced unprecedented commercial growth in recent years, particularly following the success of Netflix's *Drive to Survive* documentary series, which has broadened its global appeal and brought new audiences to the sport (Stern, 2025). In 2023, F1 reported \$3.2 billion in revenues, and in 2024, more than six million spectators attended one of 24 events across five continents (Liberty Media, 2024). Each race requires extensive operational logistics, including the complicated transport of cars and support equipment over land, air, and sea (Zinkula, 2023).

F1's global reach and operational complexity make it particularly susceptible to climate impacts, as the series navigates diverse environmental conditions while maintaining both competitive integrity and commercial viability. Thus, its race calendar must balance the preference for optimal weather conditions with economic factors, including

Table 1. Race sites in Formula 1 series at time of writing.

Location	Race circuit	Month of race (2024 season)	Contracted through
Bahrain – Sakir	Bahrain International Circuit	March	2036
Saudi Arabia – Jeddah	Jeddah Corniche Circuit	March	2031
Australia – Melbourne	Albert Park Circuit	March/April	2037
Japan – Suzuka	Suzuka International Racing Course	April	2029
China – Shanghai	Shanghai International Circuit	April	2025
USA – Miami	Miami International Autodrome	May	2031
Italy – Imola	Autodromo Internazionale Enzo e Dino Ferrari	May	2025
Monaco	Monte Carlo Harbour	May/June	2025
Canada – Montreal	Circuit Gilles Villeneuve	June	2031
Spain – Barcelona	Circuit de Catalunya	June	2026
Austria – Spielberg	Red Bull Ring	June/July	2030
UK – Silverstone	Silverstone Circuit	July	2034
Hungary – Budapest	Hungaroring	July	2032
Belgium – Spa	Circuit de Spa-Francorchamps	July	2025
Holland – Zandvoort	Circuit Park Zandvoort	August	2025
Italy – Monza	Monza	August/September	2025
Azerbaijan – Baku	Baku City Circuit	September	2026
Singapore	Marina Bay Street Circuit	September	2028
USA – Austin	Circuit of the Americas	October	2026
Mexico – Mexico City	Autodromo Hermanos Rodriguez	October	2025
Brazil – Sao Paulo	Autodromo Jose Carlos Pace	November	2025
USA – Las Vegas	Las Vegas Street Circuit	November	2032
Qatar – Lusail	Lusail International Circuit	November	2032
UAE – Abu Dhabi	Yas Marina Circuit	December	2030
South Africa – Johannesburg	Kyalami Grand Prix Circuit	(not on current calendar)	

competing events in each market and the logistics of shipping and travel. [Table 1](#) presents the 2024 race calendar, including local seasonal context for each event.

Strategic scheduling helps explain why cold weather poses minimal risk to current F1 operations, while heat remains a persistent concern despite efforts to schedule around it. For instance, while races in the Middle East are scheduled during relatively milder seasons, increasing temperatures in these regions can still push conditions beyond safe thresholds. Traditional scheduling strategies may become less effective as climate change disrupts historical seasonal patterns, potentially requiring more fundamental changes to the racing calendar in the future.

Materials and methods

We employed a multi-phase approach to establish and apply climate boundary conditions for motorsport. First, we conducted a targeted review of literature across environmental physiology, sport science, and climate impact assessment to identify potential boundary conditions. This was supplemented by semi-structured interviews with motorsport professionals to validate and refine these boundaries. We then collected and analysed historical weather data, air quality records, wildfire projections and flood risk projections for 25 current or proposed F1 circuits to assess their vulnerability against these conditions. The following sections detail our approach to boundary setting, data collection, and analysis.

Setting boundary conditions

Despite limited research on climate change and its direct impact on sport events, scholars have examined some boundary conditions under which sport participation is considered safe (e.g. maximum temperature for summer outdoor participation, safe air quality; Orr, 2021). These boundary conditions have significant implications for sport policy and warrant further investigation. The literature informing boundary conditions is interdisciplinary, spanning environmental physiology (e.g. heat, humidity, cold, and air quality on health and well-being), sport science (e.g. sport performance), engineering (e.g. playing surfaces, equipment), and management (e.g. supply chain, consumer behaviour). Thus, in [Figure 1](#), we propose a simplified framework for setting climate boundary conditions for sport (the boundaries that define *sensitivity*, which in turn inform the overall assessment of climate risks for sport organisations).

Climate boundary conditions must be established for each possible hazard (e.g. extreme heat, precipitation or storm activity, flooding, wildfire) and based on at least four key considerations listed in our framework. They include the impact on:

1. **human health**, inclusive of mental health, of athletes, team staff, officials, venue and event staff, broadcasters, volunteers, fans, and anybody else on-site or otherwise involved;
2. **sport performance**, which may include speed, strength, agility, coordination, and cognitive abilities of players and officials, among other considerations;
3. **facilities and equipment**, including the buildings, playing surfaces, access to water and electricity, entry points (e.g. parking lots, roadways, loading docks), the ball, car, bike, boat, or other equipment, and sportswear (e.g. shoes or cleats); and

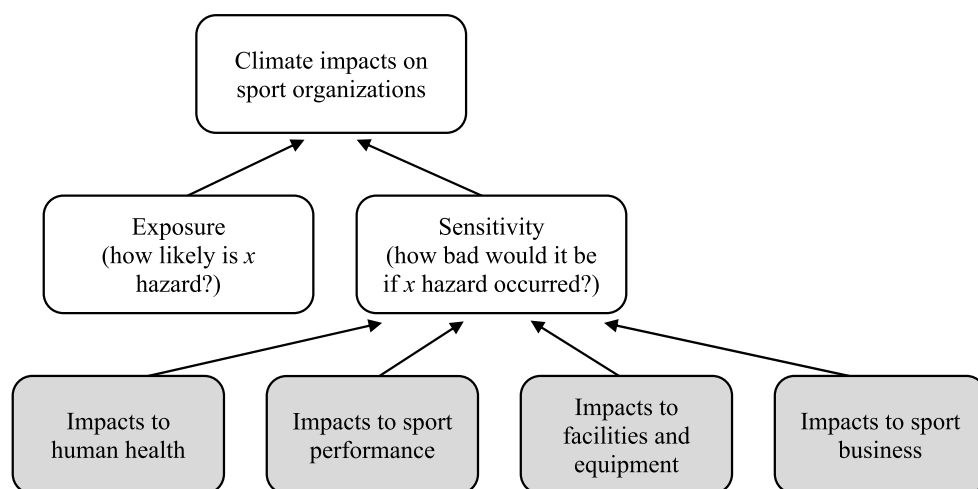


Figure 1. A simplified framework for setting climate boundary conditions in sport.

4. **sport business**, including operations, finances, consumer behaviour, and communication.

To establish a climate boundary condition, targeted reviews of the literature can generate initial starting points for each consideration (Ross & Orr, 2022). Interviews and consultations with athletes, coaches, trainers, facility and event managers, and league officials also provided valuable insights into racing organizations' past climate risks and the perceptions of those impacted. Climate boundary conditions differ based on sport, level of play, geography, risk orientation, and many other factors. For instance, the Extreme Heat Policy written by Sports Medicine Australia (2021) sets different boundary conditions for different sports since the equipment worn, type of playing surface, duration of exposure, and athletes' conditioning level will influence how heat impacts the body. This heat policy is based only on human health considerations; therefore, we suggest it be considered alongside other considerations (performance, facilities and infrastructure, and business).

As research on climate risks in sport continues to develop and more becomes known, we expect climate boundary conditions to evolve. However, their utility as a tool for assessing climate risk cannot be understated. Climate boundary conditions allow for scholars to enter into discourse with other disciplines whose research brings invaluable insight to sport management. For instance, physiologists can contribute knowledge on how the body handles different temperatures (Tyler et al., 2016) and air quality conditions (Rundell, 2012), engineers can indicate the upper limits of a stadium roof's stability under windy conditions (Biagini et al., 2007) or the drainage capacity of a sports field (Gezici et al., 2021), and management scholars might contribute to game attendance and ticket sales data based on different climate conditions (Watanabe et al., 2019), all of which is relevant to an overall assessment of climate risks.

This study was commissioned by McLaren Racing and thus, some of the methodologies described below were informed by the specific requests of the racing organization.

Setting boundary conditions for any sport remains an imprecise science, relying on interdisciplinary perspectives and inputs. What we describe are new, exploratory, and experimental boundaries; we do not propose that these boundaries are definitive or exhaustive, rather, we advance them as a starting point for further research and refinement.

We reviewed the extant literature on temperature, precipitation, air quality, surface flooding, and extreme events on elite athletes' bodies (especially in motorsport) and bodies generally (for staff, spectators, and volunteers). We considered the competitiveness of the race, including drivers' ability to stay focused (cognitive function) and the ability of the car to manoeuvre safely. We also considered the impact of these same hazards on the car (e.g. the tires, the engine), race operations (i.e. the garages, the track itself, event logistics, communication networks), and broader business interests (i.e. the fan experience, sponsor revenues, and the global logistics associated with operating a team). Additionally, semi-structured interviews were conducted with a sample of team medical doctors, physiotherapists working with drivers, team technicians, a race engineer, and retired drivers ($n=8$) to glean professional insights on existing – and potential future – risks. Participants were selected through purposive sampling, with motorsport experts identified based on expertise relative to the four key considerations in our framework (human health, sport performance, facilities and equipment, and sport business). Eleven people were initially contacted based on LinkedIn profiles that demonstrated work experience in the areas of interest, seven agreed to participate, and one additional interview was secured via snowball sampling. Interviews were conducted over Microsoft Teams in the spring of 2023 with a protocol approved by the research ethics board at Loughborough University. Participants were asked to reflect on climate hazards impacting the series in the past, including how adaptations were undertaken and what the current limits to adaptation might be. They were also asked about their concerns related to climate change generally, or specific hazards. Interviews were then transcribed verbatim and transcripts were deductively coded using NVivo software for themes relating to each of the hazards identified by the research team (e.g. heat, cold, precipitation) and each of the four areas of concern outlined in our framework (impacts to health, sport performance, facilities and equipment, and business). The findings from these interviews helped identify additional relevant literature and were used to triangulate our proposed boundary conditions, which are summarized in [Table 2](#) and detailed below.

For each condition, we ascribe two elevated levels of risk: moderate and high. This is consistent with boundary conditions in heat policies for sport which identify trigger points at several temperature thresholds (Sport Medicine Australia, 2021). Adopting the same parlance as motorsport, we symbolize moderate risks with a yellow flag and high risks with a red flag. In established F1 racing protocols, a yellow flag is employed to signal caution to the driver, while a red flag is waved to stop the race 'with reasons ranging from a serious incident to poor weather conditions' (Seymour, 2023, para. 16).

Climate boundary conditions for heat

The best environmental measure for heat and humidity would be the wet-bulb globe temperature (WBGT), a measure of ambient temperature, humidity, windspeed, sun angle and cloud cover in direct sunlight (Racinais et al., 2015); however, for pragmatic reasons we had to simplify the data collection by using ambient temperature as a

Hazard	Climate boundary condition	References
Heat	• 25.5 °C (78 °F) signifies a moderate risk based on the increased possibility of exertional heat-related illness for drivers and pit crews • 27.8 °C (82 °F) represents a high risk based on the increased possibility of exertional heat-related illness for drivers and pit crews, and heat illness (non-exertional) for fans and others on-site	Roberts & Nicholson, 2010; Smith et al., 2016
Cold	• <5 °C (41 °F) signifies a moderate risk based on health considerations, namely risk of hypothermia • <0 °C (32 °F) signifies a high risk based on risk of ice developing on the track and access points to the site	Gießler et al., 2010; Fudge et al., 2015; Riehm et al., 2012
Air quality	• The AQI's 'Unhealthy for Sensitive Groups' threshold presents a moderate risk based on the health and performance risks for drivers and pit crews • The AQI's 'Unhealthy, Very Unhealthy, or Hazardous' levels are considered high risk due to the health risks they present for all people at a race site	Air Quality Index Reporting, 1999; Archsmith et al., 2018; Heintz et al., 2022; Hung et al., 2023; Manisalidis et al., 2020
Precipitation	• >4 mm per hour signifies a moderate risk based on reduced visibility and possibility of producing pools of standing water on the track • >7.6 mm per hour signifies a high risk based on the upper limits of what the F1 rain tires can dissipate	Hautière et al., 2009; Konstantopoulos et al., 2010; Met Office, n.d.; Pirelli, n.d.
Flooding	• Moderate risk to a site if it had any experience of flooding in the last 30 years and/or under 2.0 m sea-level rise per a NASA report • High-risk was based on an NOAA threshold of 0.3 m sea level rise • Data pulled from flood maps available via Climate Central and National Centers for Environmental Information	Hooijer & Vernimmen, 2021; Sweet et al., 2022
Other extreme events	• Any recent occurrence (past 30 years) of wildfire or wildfire smoke within 200 km of the race site would present a moderate risk • Fire-prone areas based on the Fire Weather Index (by the Global Fire Weather Database) were deemed high risk. • Thresholds for fire are based on the potential for fires to lead to race cancellations or postponements due to the health concerns associated with smoke for both participants and spectators, reduced air quality, reduced visibility, and the potential for wildfires to limit access and roadways (due to evacuation routes and emergency services needs). • Any history of sandstorm (within 50 km) of the race site. • Any landslides at or near (within 50 km) of the race site. • IPCC projections for sea level rise along the nearby coasts (if the site is coastal) • Any periods of drought in the past 10 years.	NASA, n.d.; UNEP, 2022 New for this study

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Heat	• 25.5 °C (78 °F) signifies a moderate risk based on the increased possibility of exertional heat-related illness for drivers and pit crews • 27.8 °C (82 °F) represents a high risk based on the increased possibility of exertional heat-related illness for drivers and pit crews, and heat illness (non-exertional) for fans and others on-site	Roberts & Nicholson, 2010; Smith et al., 2016
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substitute, since we were collecting historical weather data which generally does not include information on wind speed, sun angle or cloud cover, and only occasionally includes humidity. It is important to note that the effects of heat on athletes can vary widely due to differences in sports-related factors such as clothing, equipment, exposure duration, play duration, and the nature of the sport itself (Ross & Orr, 2022). Therefore, the measure of ambient temperature (WBGT) in a best case-scenario, or simply ambient temp in this case must be considered alongside clothing, equipment (e.g. helmets being worn), and metabolic heat production.

The physiological demands of F1 racing make heat exposure particularly challenging. Drivers compete in fireproof suits and helmets while seated in open cockpits where temperatures can reach 60 °C during races (Cooper, 2023). During a race, drivers can up to 4.5 kg in body weight through sweating (Hosie, 2022). In recognition of these challenges, the FIA has begun implementing mandatory driver cooling systems, though these mitigations have limitations (Benson, 2024). Recent events, such as the 2023 Qatar Grand Prix where multiple drivers required medical attention, demonstrate that current heat management strategies may be insufficient as global temperatures rise. Furthermore, in the case of road-based competitions such as motorsport, road conditions such as the type of surface (e.g. dark asphalt, reflective pavement, or shaded asphalt) can affect the progression of heat strain in athletes (Vanos et al., 2024). Taking all this into consideration, for the temperature boundary we adopt the following: 25.5 °C (78 °F) signifies a moderate-risk level, while 27.8 °C (82 °F) represents a high-risk level (Roberts & Nicholson, 2010; Smith et al., 2016). Sports medicine specialists and kinesiologists have established these thresholds as critical points where the likelihood and severity of heat-related illnesses is high among athletes, especially endurance athletes like marathon runners and road cyclists, who compete on roadways. The Medical and Scientific Commission of the IOC and several international federations have suggested that event organisers monitor and communicate the environmental conditions before and throughout the event (Racinais et al., 2015). The levels of performance impairment and health risk for a given environment depend on the heat production of the sport and its heat dissipation capacity, as well as athlete characteristics (e.g. body size and heat acclimation status).

Climate boundary conditions for cold

Climate change also stands to bring more flash freeze events as weather events shift the polar vortex away from the poles, as seen in recent freezes in the midwestern United States and Texas (Ranieri, 2024). Cold temperatures present a risk of hypothermia and ice forming on the track, both of which can lead to catastrophic outcomes. The risk of hypothermia becomes a concern at temperatures of approximately 41 °F (5 °C; Fudge et al., 2015). However, in motorsport, where individuals are either in full racing suits (drivers and pit crew) or have the option to layer clothing (staff, volunteers and spectators), the risk of hypothermia is relatively low. We also note that the F1 calendar currently chases the sun, traveling to warm weather locations in their respective summer seasons, thus cold weather was not a significant concern in the series' current format.

The primary concern is the formation of ice on the track surface, which generally occurs around the freezing point of water (Riehm et al., 2012). Nevertheless, due to the heat generated from tire friction on the track during racing, it is conceivable that the freezing point on a racetrack may be lower; this possibility is, however, an aspect

that remains unclear and unexplored, lacking comprehensive studies (Gießler et al., 2010). While cold temperatures have not yet been an issue on the race calendar, recent instances of rapid freeze events in regions like Western Europe, the United States, the Middle East, and China necessitate vigilance in monitoring cold-related risks.

Climate boundary conditions for air quality

Reliable air quality guidelines are unavailable in sport-related scholarship. Therefore, we utilize the Air Quality Index (AQI), which is the global scientific standard for reporting air pollution levels (Tsegaw & Alemayehu, 2019). AQI is a simplified tool for assessing air quality since various air pollutants have differing characteristics and degrees of harm. To make this assessment more straightforward, the U.S. Environmental Protection Agency (1999) introduced the Pollution Standard Index (PSI), which categorizes AQI readings based on the potential health effects of different levels of air pollution. The PSI comprises six levels, classified as follows:

1. Good: $AQI \leq 50$
2. Moderate: AQI 51–100
3. Unhealthy for Sensitive Groups (individuals with underlying respiratory conditions): AQI 101–150
4. Unhealthy: AQI 151–200
5. Very Unhealthy: AQI 201–300
6. Hazardous: $AQI > 300$

These classifications have been endorsed by the World Health Organization as the standard for assessing air quality concerning health, including its relevance to sporting activities (World Health Organization, 2006). For our analysis, we consider the ‘Unhealthy for Sensitive Groups’ threshold to present a moderate risk because of the various personnel working in the pit areas. In many cases, these crew members will be physically exerting themselves and taking in higher volumes of air, thereby increasing their exposure to air pollution. Moreover, because of their proximity to the track and cars, crew members are more likely to be directly exposed to emissions from race cars and other vehicles. As such, AQI levels in the Unhealthy, Very Unhealthy, or Hazardous levels are considered high risk for all present at an event.

Climate boundary conditions for precipitation

Based on the assumptions that (1) any measure above 4 mm in less than an hour (the Met Office’s threshold for heavy rain) could have deleterious impacts on visibility and risks producing pools of standing water on the track and (2) anything above 7.6 mm in less than an hour (IPCC’s definition of heavy rain) would be pushing the upper limits of what the most effective tires (i.e. full wet) can dissipate, we adopt the following thresholds. First, a yellow flag is assigned to any site receiving 4–7.9 mm of rain per hour or greater, while a red flag is assigned to any site receiving 8 or more mm of rain per hour.

Considering that rainfall is not usually consistent across a whole day, for daily figures, we conservatively set the yellow flag threshold at 36 mm in a day (up to 8 hr in the day at the 4 mm moderate-risk level) and the red flag at 60.8 mm in a day (up to 8 hr in the day

at the 7.6 mm high-risk level). We acknowledge limitations with these thresholds; a fast-moving storm could roll over a track, drop 7.6 mm of rain in one hour, and move on, and this case would not be picked up in this analysis. We also acknowledge that the drainage capacity of a track or race course will radically alter its capacity to cope with rain, and that the FIA has requirements for drainage on track – specifically that tracks have a transversal incline of 1.5% to 3% on straights to drain water (measured either from one side of the track to the other, or from a midpoint to each side; Benedetto et al., 2012; Kmoniček et al., 2019). However, it was not possible to identify the specific drainage capacity of all tracks with any publicly available sources, and based on recent rain events during races, it is clear some tracks continue to pool water in certain places. Future research ought to measure the drainage capacity of all tracks across the FIA’s sanctioned venues which could assist in future boundary-setting evaluations.

Climate boundary conditions for flooding

Unlike pluvial flooding – the pooling of water on land caused by excess rain that cannot be absorbed into the soil – river or coastal flooding is caused by the overflow of bodies of water like rivers, lakes, or oceans. Coastal flooding has impacted roadways and coastal infrastructure globally for several years.

For tracks on or near the coastline, coastal flooding due to sea level rise or king tides during storms is a risk factor worth monitoring. King tide-related coastal flooding can lead to inundation or erosion, saltwater intrusion, and disruption to daily life and economic activities. Flooding from rising sea levels does not imply permanent inundation but could mean increased risk from the temporary flood events. As coastal flooding can interrupt traffic and other movements in an area by shutting down roadways and airports, it may also affect the access routes to the race. We determined the high-risk threshold based on the 2022 Sea Level Rise Technical Report from the United States’ National Oceanic and Atmospheric Administration which used 0.3 m (1.0 ft) of sea level rise and a moderate risk based on Hooijer and Vernimmen’s (2021) use of a 2.0-m sea level rise threshold.

Riverine flooding may present similar issues when rivers overflow and burst their banks, typically the result of a combination of heavy rain and already-high water levels. We flagged any sites (1) in which flooding occurred in the past decade or (2) deemed to be at elevated risk of flooding in the coming three decades. We consider any history of flooding at the race site to be a cause for concern and flooding nearby (impacting access roads or critical infrastructure such as airports and hospitals) to present a moderate risk.

Other climate boundary conditions

Other extreme events can significantly impact motorsport races. First, wildfires or bushfires can occur in various parts of the world, but they are most associated with regions that experience hot and dry climates (Arrogante-Funes et al., 2022). Key risk factors for wildfires include dry weather, high temperatures, strong winds, and the presence of flammable vegetation (United Nations Environment Programme, 2022). Wildfires may lead to race cancellations or postponements due to safety concerns for both participants and spectators. Poor air quality caused by wildfire smoke can also pose health risks to athletes and attendees, affecting their performance and well-being.

In this study, the climate boundary condition for wildfire was any recent occurrence (i.e. past 30 years) of wildfire or wildfire smoke within 200 km of the race site.

Second, sandstorms can occur in arid and desert regions. Risk factors for sandstorms include strong winds, dry conditions, and loose soil or sand in the area (Opp et al., 2021). Sandstorms can disrupt racing events by reducing visibility, creating hazardous driving conditions, and causing damage to equipment. They can also impact the health and well-being of people on-site, primarily regarding discomfort, sore eyes from sand exposure, and cardiovascular and respiratory challenges, particularly for those with underlying conditions (Tobias et al., 2019). These conditions may lead to race delays or cancellations. The climate boundary condition for sandstorms was any history of sandstorms, as these require proximity to dunes, deserts, or vast deposits of sand.

Third, landslides and mudslides can occur in regions with steep terrain and heavy rainfall. High-risk areas include mountainous regions, coastal cliffs, and areas prone to monsoons or heavy rainfall (Davies, 2015). Key risk factors include heavy and prolonged rainfall, rapid snowmelt, soil erosion, and unstable geological conditions (Sim et al., 2022). Landslides and mudslides can pose serious threats to racing events, especially if they occur along race routes or near venues. These events can lead to road closures, race postponements, or even accidents if participants or spectators are caught in the path of a landslide or mudslide. The climate boundary condition for landslides and mudslides was any history of landslide or mudslide at or near the race site or critical event-related infrastructure (e.g. the international airport which would be the entry and departure point for all athletes, team staff, and media members), combined with slope gradient data, vegetation data, and precipitation data (i.e. too much or too little water can result in soil structure imbalances which can lead to mudslides or landslides; Dai et al., 2022).

Finally, droughts can affect regions worldwide with varying degrees of severity. Droughts are primarily driven by a prolonged lack of rainfall (Hedayat & Kaboli, 2024) and can impact racing events in multiple ways. Water scarcity can affect track maintenance and facility operations, while drought conditions may lead to stricter water-use regulations, thus affecting hospitality and event management. In extreme cases, racing events in drought-affected areas may need to be relocated or postponed due to safety concerns, especially if fire risk is also elevated during the drought. The climate boundary condition for drought was a history of drought or a predicted significant reduction in precipitation due to climate change.

Data collection

Data collection centred on 25 cities. Twenty-four cities corresponded to the 24 circuits appearing in the 2024 FIA F1 World Championship race calendar. Johannesburg was included because of ongoing efforts by the South African government to bring F1 racing back to the country after a three-decade absence (Vecchiatto, 2024). For temperature and precipitation data, we used the National Centers for Environmental Information's (NCEI) Climate Data Online databases to draw historical weather records from weather stations nearest each race circuit. Daily weather records dating to 2013 were collected and analysed based on the number of days in each month the temperatures reached the moderate- or high-risk thresholds. We collected data from 2013 onward because this is the timeframe for which data was consistently and reliably available for

all race sites (with notable exceptions; e.g. the data for U.S. and European sites was available for much longer). It is worth noting that daily highs tend to be much higher than the daily averages, thus if the average is above the high-risk threshold, chances are, the daily high was several degrees warmer in the mid-day hours when most races occur. Precipitation data were coded based on the number of days in a given month over the last decade that a moderate or high-risk threshold was surpassed in a day.

For air quality, daily observations were collected from government-run monitors for each country and location. These were cross-referenced with monitors operated by the U.S. Embassy in each region, particularly where the government-supplied data were not reliable or consistent. In this case, we coded the data based on patterns of high pollutant and particulate matter counts at a given time of the year (e.g. in India, the harvest happens in October, and it is the custom to burn fields following the harvest, which produces a substantial seasonal spike in air pollution; or in Canada, the wildfire season is typically June to September). For city-based races, we also looked for regional patterns to check whether some parts of the city are cleaner than others and might be better suited to a race. A similar approach has been used to evaluate air quality during the 2024 Olympic and Paralympic Games in Paris (Bougault et al., 2024).

Flooding data were drawn from flood maps for each location (i.e. the race site itself, if available, or the regional map of the area) to determine if coastal, riverine, or pluvial flooding is a risk. Data were collected from NCEI flood maps for the U.S. locations and Climate Central flood maps for other locations, supplemented with local inspections of watersheds, river networks, and surface absorption rates on GIS (e.g. concrete is not absorptive, but green space is; in that case, soil and surface cover data are checked).

For the remaining climate events, we reviewed historical records in each race location to screen for any major incidents in the past decade. Regarding wildfire risk, we also analysed regional statistics in the Global Fire Weather Database (NASA, n.d.). Finally, we reviewed the IPCC 6th Assessment Report (2023) to cross-reference our findings against predictable changes in weather patterns, sea level rise, and land use change in the coming decades. Table 3 presents data sources and analytical approaches.

Findings

Table 4 lists all risks for each race, coded for moderate or high risk relative to each hazard. Overall, our findings show that extreme heat is the most common and physiologically significant hazard threatening the current F1 calendar, impacting 19 races on the circuit (moderate risk in 11, high risk in 8), while flooding could impact 18 race sites (high risk in 1, moderate risk in 17), and poor air quality could reduce breathability at 10 sites (moderate risk in 9, high risk in 1). Heavy rains are also a concern at seven locations (all moderate risk). Five races are also at risk of being impacted by wildfires based on their Fire Weather Index scores (moderate risk in 4, high risk in 1), while Melbourne is at moderate risk due to its recent history of wildfires (i.e. the 2020 fire season impacted most of Victoria).

The results demonstrate that many circuits are experiencing increasing temperatures, with venues like Budapest, Austin, and Las Vegas facing a growing number of days above the high-risk threshold of 27.8 °C (82 °F). Flooding poses a significant threat to several tracks, with Saudi Arabia's Jeddah circuit at critical risk from even minor sea level rise

Table 3. Sources of data and points of analysis.

Types of data	Source(s) of data	Analysis
Temperature (Heat and cold)	National Centers for Environmental Information (NCEI)	Review historical weather records (dating to 2013) to check for instances where ambient daytime highs topped moderate-risk thresholds of 25.5 °C (78 °F) or high-risk thresholds of 27.8 °C (82 °F).
Air quality	Each government's database, and U.S. Embassy-run monitors where government data was unreliable	Review historic weather records (dating to 2013) to check for instances of 'unhealthy' or 'hazardous' AQI (following EPA standards)
Precipitation	NCEI	Review historic weather records (dating to 2013) to check for instances where 36 mm was surpassed in a day (up to 8 hr in the day at the 4 mm moderate-risk level), or that a high-risk threshold of 60.8 mm was reached in a day (up to 8 hr in the day at the 7.6 mm high-risk level).
Flooding	NCEI and Climate Central Maps	Flooding caused by sea level rise under 0.3 and 2.0 m and review of historic records of flooding
Other extreme events	News databases and government reports for each country Global Fire Weather Database (GFWED)	Flagged any recent extreme weather event, king tide, land shift (e.g. landslide, mudslide), wildfire, drought, etc. Global Fire Weather Index ratings for each site, sourced on GFWED website.
Future projections	IPCC report	Borrowing models from IPCC 6th Assessment, we checked for signals of future change to weather patterns and air quality, and sea level rise.

and Singapore at a marginally higher elevation. Additionally, air quality is a significant concern for Doha (consistently records AQI levels above 100), Mexico City and São Paulo (significant fluctuations in air quality), and Johannesburg (hazardous AQI levels exceeding 400 in some instances).

The most at-risk races overall are Singapore and Qatar. Singapore faces multiple threats related to heat, precipitation, flooding, and air quality. The city – state is at critical risk from sea level rise, as about 30 percent of its land is less than 5 m above sea level (National Climate Change Secretariat, 2024). It also experiences sporadic high-risk temperature levels throughout the year and is prone to heavy rainfall events. Elsewhere, Qatar is challenged by extreme heat and poor air quality. Temperatures above the high-risk threshold are typical from May through October, and the area consistently records high AQI levels, with October often seeing average AQI values of 175 and above.

In contrast, some circuits face fewer climate risks. The Red Bull Ring in Spielberg, Austria, is one of the least at-risk venues. It experiences infrequent moderate-risk heat levels, has very low AQI levels (mostly below 50), and faces minimal flood risk. Still, it is important to note that even these relatively safer venues are not immune to the effects of climate change. Many sites are projected to face significantly more frequent and intense heat waves in the coming decades, potentially making summer races untenable by mid-century.

Discussion

Implications for Formula 1 and motorsport

The findings offer a detailed climate risk assessment for F1 races while providing contributions to both research and practice more broadly. All circuits examined in this analysis are projected to experience the impacts of climate change except Spielberg. This finding

Table 4. Summary of climate risks by race (in month of race in 2024).

Race	Extreme heat	Cold	Air quality	Precipitation (too much precipitation)	Flooding	Other extreme events
Bahrain – Sakir	Moderate		Moderate			
Saudi Arabia – Jeddah			Moderate		High	
Australia – Melbourne	Moderate		Moderate		Moderate	Moderate ^a
Azerbaijan – Baku					Moderate	Moderate ^{a,b}
USA – Miami	High			Moderate	Moderate	
Italy – Imola	Moderate		Moderate	Moderate	Moderate	
Monaco					Moderate	
Spain – Barcelona	High					
Canada – Montreal			Moderate		Moderate	
Austria – Spielberg						
UK – Silverstone	Moderate				Moderate	
Hungary – Budapest	High				Moderate	
Belgium – Spa	Moderate			Moderate	Moderate	
Netherlands – Zandvoort	Moderate				Moderate	
Italy – Monza	High				Moderate	
Singapore	High		Moderate	Moderate	Moderate	
Japan – Suzuka	Moderate			Moderate	Moderate	Moderate ^b
Qatar – Lusail	High		High			
USA – Austin	Moderate				Moderate	Moderate ^a
Mexico – Mexico City						Moderate ^c
Brazil – Sao Paulo	Moderate		Moderate		Moderate	Moderate ^a
USA – Las Vegas	Moderate				Moderate	High ^{a,c}
UAE – Abu Dhabi	High				Moderate	
China – Shanghai	High		Moderate	Moderate		
South Africa – Johannesburg	Moderate		Moderate	Moderate		

^aWildfire.^bLandslide.^cDrought.

should serve as an alarm for this global sport: to secure sustainable outcomes both environmentally and economically, race officials will need to plan accordingly for future editions of the series. The risks that F1 faces are broad, covering all potential risks identified except cold while also being specific to each site (see Table 4; cf. Schneider & Mücke, 2021). Each circuit has its own unique set of environmental conditions that differentiate its climate vulnerabilities. For a sport that competes globally, the uniqueness of each site is problematic as there is no one-size-fits-all solution for these climate vulnerabilities across all locations. This issue is compounded by the fact that each circuit and Grand Prix is organised individually rather than centrally by F1 or its governing body, the FIA.

Climate change projections suggest that some races may become untenable if adaptations do not occur or more serious global efforts are made, an outcome similar to what is predicted for the Winter Olympic and Paralympic Games (Scott et al., 2015). Beyond F1, motorsport at large is at risk as many other racing series (e.g. Formula 2, Formula E, Grand Prix motorcycle racing) also make use of these same circuits. Thus, risk management and adaptation may be shared among all professional racing series, as there is no distinct advantage to keeping climate solutions to themselves. However, we acknowledge that some circuits without F1 Grand Prix events would like to be added to the calendar in the future (i.e. South Africa).

Adaptations are needed to reduce risk while ensuring the safety of athletes, officials, and spectators. For the most common risks found (i.e. heat, flooding, and poor air

quality), these adaptations can vary from simple additions to operations to more extreme changes to the sport itself. More medical support could be brought on-site to support acute health issues through cooling measures and KN95 masks. Car and tire manufacturers may need to consider a broader range of conditions when developing their respective components (e.g. wet tires may need to better disperse water than previously). Racing rules may need to include adaptations seen in football, like drinking and cooling breaks at specific temperature thresholds, athlete and crew temperature checks, or a temperature limit that would lead to a red flag. A more extreme suggestion for adaptation would be to reorganise the racing calendar entirely to ensure that each location is optimized for its climate conditions while disregarding the racing calendar traditions currently upheld (Orr, Murfree, et al., 2022).

Theoretical and methodological implications

This study describes an initial effort to set boundary conditions for motorsport, using the F1 series as a testing ground. In doing so, it contributes to research on the climate impact on organisations concept, or CIO, originally developed by Hinkel (2011) and brought to sport organisations via Orr and Inoue (2019), by applying the concept to motorsport organizations (specifically, from the vantage point of competing teams and race organizers) and finding the current limits of available data and knowledge (e.g. no historical weather data that includes WBGT, limited data on track drainage). While previous sport ecology research has examined contexts like the Olympic Games and World Cups more extensively (Ross & Orr, 2022; Scott et al., 2015), this study is the first to focus on climate vulnerability in motorsport. F1 spans the entire globe in a single season and is regularly exposed to a diverse variety of climates – and therefore climate risks – but it can also rapidly experience such changes on a weekly basis. Thus, F1's status as a 'world tour', coupled with its rapidly growing popularity (Smith, 2024), makes it a unique and complex context for study. If a climate risk assessment is possible within the context of F1, it should be replicable and useful for a variety of other (potentially more geographically confined) sport contexts.

The framework for setting boundary conditions described in this manuscript should serve as a foundation for future research in climate risk assessment for sport events. Still, it is not immune to constructive critique. The boundary conditions were developed based on other sports and contexts with input from industry personnel to adapt them to motorsport (as summarized in Table 2). Thus, the present boundary conditions are specific to motorsport and should be refined through future iterations of research and other academic disciplines (e.g. medicine, engineering, business). Given the unique nature of each sport and community (Schneider & Mücke, 2021), a fit-for-purpose climate risk analysis that is readily available to those who need the support serves the needs of academics and industry practitioners alike.

Future research

As this work involves prediction, it is important to emphasize that prediction work can be imprecise (Pedersen et al., 2021). In some cases, the statistical likelihood of a heavy rain event or a scorching heat event may have been extremely low. Yet, its past

occurrence suggested the possibility of it occurring again. Therefore, these findings should be viewed as a guide that sets expectations while not necessarily covering every possibility to perfection. While modelling provides some indication of likely outcomes, climate change has accelerated the likelihood of flash flooding from heavy rain events, heat waves, and flash freeze events in some regions. Thus, it is impossible to predict the future with certainty, and predictions will have to evolve as we learn more about the pace of climate change in the years ahead.

We also note that the present study primarily focuses on assessing climate risks and their potential impacts on a sport (i.e. F1) without considering the sport's ability to meet these environmental challenges. This approach is consistent with the CIO element of Orr and Inoue's (2019) CVSO framework. As discussed previously, the CVSO framework can be applied to analyse (1) the potential impacts of climate change on organisations and (2) their capacity to adapt to these impacts. While our research thoroughly examines the CIO aspect by detailing potential climate hazards for F1 venues, it does not address the OCC element, which represents an organisation's ability to adapt to and cope with these climate impacts. Future research should explore the OCC of F1 stakeholders (teams, media, the FIA) and the broader motorsport industry to provide more robust insights into adaptation potential and how such adaptations might mitigate the climate risks. While we have offered some potential adaptations in this discussion, their feasibility and effectiveness remain unknown. Expanding research to encompass both CIO and OCC would further bridge the gap between academic knowledge and practical insights, assisting industry personnel in navigating the challenges posed by climate change.

While not the focus of this research, we acknowledge that F1's global operations contribute significantly to climate change through carbon emissions from both racing operations and logistics (Sturm et al., 2025). The series format, requiring extensive international travel and generating considerable emissions, presents challenges for alignment with international climate targets. Indeed, we recognize the unfortunate irony that motorsport's contribution to climate change may exacerbate the very environmental challenges identified in this study. While F1 has announced plans to achieve net-zero carbon emissions by 2030 (FIA, 2024), the feasibility of maintaining a global racing calendar while meeting international climate commitments remains an important area for future research.

Conclusion

Climate change is presenting a new set of challenges to the fast-paced world of motorsport. For F1, which travels to five continents and extends from March to December, it is increasingly difficult to avoid the impacts of shifting weather patterns, landslides, fires, floods, and air pollution. Our findings indicate that some events are more at-risk than others, with Singapore and Qatar facing multiple high-risk challenges. At the same time, extreme heat emerges as the most common challenge for the series, impacting 19 destinations. The diversity of risks across different locations underscores the complexity of addressing climate change in a global sport. It also highlights the need for tailored strategies that consider the unique environmental conditions of each circuit.

On the other hand, F1 is a series defined by rapid innovation and significant investment. Thus, there is potential for race organisers and teams to introduce adaptations that would benefit not just the F1 series but others that follow their lead, including other motorsport series and the car manufacturing sector overall (Autovista Group, 2022). These adaptations could range from technological innovations in vehicle design to changes in race scheduling and venue selection. As our understanding of climate risks continues to evolve, so must the strategies to mitigate them, ensuring the longevity and safety of this global sport. The challenge of climate adaptation in F1 presents an opportunity for the sport to demonstrate leadership in sustainability and resilience, potentially influencing practices far beyond the racetrack.

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